

Benchmark

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

Benchmark Clarifications:

Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.

Clarification 2: Instruction includes recognizing that these transformations preserve angle measure.

Connecting Benchmarks/Horizontal Alignment

- MA.912.GR.1.2, MA.912.GR.1.6
- MA.912.GR.3.2, MA.912.GR.3.3
- MA.912.GR.6.5

Terms from the K-12 Glossary

- Coordinate Plane
- Dilation
- Origin
- Reflection
- Rigid Transformation
- Rotation
- Scale Factor
- Translation

Vertical Alignment

Previous Benchmarks

- MA.8.GR.2

Next Benchmarks

- MA.912.F.2.2, MA.912.F.2.3, MA.912.F.2.5

Purpose and Instructional Strategies

In grade 8, students were introduced to transformations and whether they preserve congruence or similarity. In Geometry, students determine which transformations preserve distance (rigid transformations, or rigid motions) and which do not (non-rigid transformations, or non-rigid motions), while understanding that all the transformations in this benchmark preserve angle measures. This discussion leads to the definition of congruence and the definition of similarity in terms of rigid and non-rigid motions. (*MTR.3.1, MTR.5.1*) In later courses, other non-rigid motions are studied, including stretches in one coordinate direction.

- Instruction includes the comparison of a variety of geometric figures before and after a single transformation (including reflections, translations, rotations and dilations) to solidify that each of these transformations preserves angle measure. (*MTR.4.1*)
- Instruction includes the student understanding that in order for a transformation based on stretching and shrinking to preserve angle measure, it must have a stretch or shrink of the same scale factor in both the x -direction and the y -direction.
 - For example, the transformation $(x, y) \rightarrow (2.5x, 2.5y)$ would preserve angle measure, but the transformation $(x, y) \rightarrow (2.5x, 3.5y)$ would not.
- While the intent of the benchmark is to focus on dilations, instruction includes the introduction of other non-rigid transformations to showcase that not all non-rigid transformations preserve angle measure.
 - For example, “stretch” or “shrink” in the direction of one of the axes, such as $(x, y) \rightarrow (5x, y)$ or $(x, y) \rightarrow \left(x, \frac{y}{6}\right)$, does not preserve angle measure. Students

can visualize this using geometric software to see how each affects angle measures of a triangle.

- Transformations should be presented using words and using coordinates.
- Instruction includes the use of folding paper (e.g., patty paper) for hands-on experiences, as needed, and interactive geometry software to allow more flexibility in exploration, when possible.

Common Misconceptions or Errors

- Students may incorrectly assume that dilations change angle measures.
- Students may assume that dilations only occur in either the x -direction or the y -direction based on knowledge of function transformations.
- Students may assume that every non-rigid motion preserves angle measures since the only non-rigid motions in the benchmark are dilations, which do preserve angle measures. But many non-rigid motions preserve neither length nor angle measure.

Instructional Tasks

Instructional Task 1 (MTR.4.1)

Penelope made the following statement in Geometry class, “Figure A is a rotation of Figure B about the origin.” Chalita disagreed because distance and angle measures are not preserved between the two figures.

- Figure A has the coordinate points $(1, -1)$, $(3, -1)$ and $(1, -2)$.
- Figure B has the coordinate points $(0.8, 1)$, $(1, 3)$ and $(2, 0.8)$.

Part A. What does “distance and angle measures are not preserved” mean in relation to the two figures?

Part B. Determine whether angle measures were preserved from Figure A to Figure B.

Part C. Determine whether distance measures were preserved from Figure A to Figure B.

Part D. Based on your answers from Part B and Part C, determine which student is correct.

Instructional Task 2 (MTR.3.1)

Sort the following transformations into preserves distance and does not preserve distance.

Counter-clockwise rotation about the origin.	A translation that moves a figure to the right and up.	A reflection over the line $x = 0$.
A dilation of $\frac{1}{2}$.	Clockwise rotation about the point $(2, -1)$.	A translation that moves a figure to the left and up.
A translation that moves a figure from quadrant I to quadrant III.	A dilation of -3 .	Reflection over the x -axis.

Instructional Items

Instructional Item 1

Circle the transformations that can be used when it is important to preserve angle measure.

Horizontal Translations	Reflections	Clockwise Rotations
Dilations	Vertical Translations	Counterclockwise Rotations

Instructional Item 2

Circle the transformations that can be used when it is important to preserve distance.

Horizontal Translations	Reflections	Clockwise Rotations
Dilations	Vertical Translations	Counterclockwise Rotations

Instructional Item 3

Write a transformation, or sequence of transformations, that preserves angle measure but does not preserve distance.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*